Fiscal Year:	FY 2012	**Task Last Updated:**	FY 12/09/2011
PI Name:	Morris, Christopher James Ph.D.		
Project Title:	Cardiovascular and Metabolic Consequences of Circadian Misalignment		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Boston	**State:**	MA
Zip Code:	02115-5804	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 NSBRI-RFA-10-01 Postdoctoral Fellowships
Start Date:	11/01/2010	**End Date:**	10/31/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Change to end date from 11/30/2012 to 10/31/2012, per NSBRI (Ed., 9/22/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Scheer, Frank (MENTOR/Brigham and Women's Hospital)		
Grant/Contract No.:	NCC 9-58-PF02303		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP Original project aims/objectives: The risk of adverse cardiac events has been listed as a major priority in the NASA Bioastronautics Roadmap (Risk Areas 5 and 6; 2010; http://bioastroroadmap.nasa.gov). 'Slam shifts', which entail rapidly changing an individual's sleep/wake cycle, are frequently (e.g., in 2,043 days of ISS operations between 2000 and 2006, slam shifts occurred on 13% of these days) performed by astronauts. Moreover, it should be noted that the effects (e.g., fatigue) of slam shifting can persist for many days after a slam shift has been undertaken. Epidemiological research indicates that slam shifting is associated with adverse endocrine, inflammatory and cardiovascular function. The possible adverse cardio-metabolic consequences of slam shifts could significantly harm an astronaut's performance and therefore the likelihood of mission success. The association between slam shift work and health problems may be due to circadian misalignment. However, the limitation of any epidemiological study is that the underlying mechanisms cannot be deduced with any degree of certainty. In particular, there is no separation of the effect of circadian misalignment per se and differences in individual characteristics, socioeconomic status, and life-style factors, such as alcohol, caffeine, nicotine, and eating habits. Thus, this calls for an experimental study to investigate the effect of circadian misalignment per se and to assess in detail potential underlying mechanisms for the increased health risk with slam shift work. Our primary aim is to determine the effect of simulated slam shift work on physiological changes that increase susceptibility to the development of cardiovascular disease. This will be accomplished by comparing cardio-metabolic, vascular, endocrine and inflammatory changes when subjects are scheduled to a 3-day simulated slam shift schedule versus a 3-day simulated day shift schedule. To address the question whether or not individuals adapt to repeated slam shift exposures, we aim to study a control group of day workers and a group of chronic slam shift workers in an identical experimental. Our independent variables include: (a) circadian alignment (misaligned [slam shift] vs. aligned [day shift]); (b) shift duration (first day vs. 3rd day on slam shift or day shift); (c) chronic slam shift workers vs. non-slam shift workers. Our dependent variables include: (a) cardiac and autonomic nervous system measures (catecholamines, sympatho-vagal balance and blood pressure); (b) peripheral vascular endothelial function (flow mediated vasodilatation, augmentation index); (c) insulin sensitivity and release measured in response to a mixed meal; (d) inflammatory markers (CRP, IL-6, TNF-alpha, MCP-1). Key findings: To date, we have successfully completed the study of two subjects that each undertook two 8-day in-laboratory stays separated by 2-12 weeks. Thus, at this stage of the project we do not have any findings to present. Impact of key findings: In the near future, it is likely that humans will return to the moon and undertake missions to Mars and other destinations in our solar system. Thus, it is vital that we understand the adverse health effects of spaceflight (e.g., circadian misalignment caused by slam shifts). This will ensure that countermeasures are developed to limit or alleviate the adverse health consequences of spaceflight and thus ensure mission success. Moreover, approximately 8.6 million Americans undertake some type of slam shift work and consequently the findings from this project will be applicable to ground-based people as well as astronauts. Research plan for the coming year: Due to study being 'up and running', we believe we can complete the study of another 12 subjects over the next year in addition to the two subjects that have already completed the study. This will be achieved by increasing recruitment effort and training more staff to work on the project.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Approximately 8.6 million Americans undertake shift work routines similar to those experienced by astronauts on the ISS. Thus, the findings regarding adverse effects of shift work on cardiovascular and metabolic factors from the current project are applicable to both astronauts and people on Earth.
Task Progress:	Across the first year of the fellowship, the protocol design and outcome measurements were finalized, the final IRB approval was obtained, and I have learned how to lead complex and multi-day human physiological studies in controlled laboratory conditions and have learned to perform all the required research techniques (e.g., indirect calorimetry). In addition, I have trained two research assistants to perform various research tasks, including subject recruitment and screening, and research measurements. As a result of the above progress, 37 people have signed our screening consent form and after extensive screening procedures (i.e., psychological and physical examinations) 2 of these subjects have successfully completed the study protocol, including four weeks of ambulatory recordings of sleep/wake cycle and two 8-day in-lab stays separated by 2-12 weeks. One subject completed the first in-lab stay but withdrew consent due to other commitments.
Bibliography Type:	Description: (Last Updated: 08/24/2020)
Articles in Peer-reviewed Journals	Morris CJ, Aeschbach D, Scheer FA. "Circadian system, sleep and endocrinology." Molecular and Cellular Endocrinology. 2011 Sep 10. [Epub ahead of print]. http://dx.doi.org/10.1016/j.mce.2011.09.003 ; PMID: 21939733 , Sep-2011